Chapter 4

Stability Analysis

4.1 Preliminary Remarks

In Chapter 3 we have established the relationship between three major paradigms of numerical analysis: Consistency, stability, and convergence. We have shown that for a consistent linear scheme, stability and consistency are equivalent. If we are faced with the task of analyzing a finite difference scheme we would proceed in the following manner: First, we would establish consistency, as well as the order of consistency. If we can also prove that the scheme is stable, convergence follows from the Lax Equivalence Theorem. Moreover, for smooth enough solutions, the order of convergence is the same as the order of consistency.

Consistency analysis of a finite difference scheme is often relatively straightforward. It involves a Taylor Series expansion, and some algebraic manipulation. We have already seen some examples. At this point, however, we have no practical way of determining the stability of a difference scheme. We present a few methods here. Perhaps the most straightforward approach is to start from the definition of stability, and use the matrix form of difference schemes that was introduced in Chapter 3. Recall from the definition of stability, Definition 3.2 that we need to estimate powers of the evolution operator $E_\Delta(\tau)$. If these can be bounded, it implies a bound on the solution, as

$$\|E_\Delta(\tau)^n u_0\| \leq \|E_\Delta(\tau)^n\| \, \|u_0\| \leq C(T) \, \|u_0\| \,, \qquad 0 \leq \tau \leq \tau_0, \quad n\tau \leq T.$$

Here $C(T)$ is the bound for $\|E_\Delta(\tau)^n\|$, which according to Definition 3.2 is allowed to depend only on T. In view of the inequality $\|E_\Delta(\tau)^n\| \leq \|E_\Delta(\tau)\|^n$, which holds for any operator norm, the condition

$$\|E_\Delta(\tau)\| \leq 1, \qquad 0 < \tau \leq \tau_0, \tag{4.1}$$

is clearly sufficient for stability. In practice, it should usually be considered necessary, as the next remark clarifies.

Remark 4.1 Condition (4.1) is *not* strictly necessary for stability. It is enough to require

$$\|E_\Delta(\tau)\| \leq 1 + \alpha\tau,$$

where α is a positive constant, independent of h or τ. From Taylor series we have the estimate $(1 + \alpha\tau) \leq e^{\alpha\tau}$. So we obtain for $n\tau \leq T$,

$$\|E_\Delta(\tau)\|^n = (1 + \alpha\tau)^n \leq e^{\alpha n\tau} = e^{\alpha T}, \tag{4.2}$$

and the stability criterion in Definition 3.2 is satisfied. However, it may be dangerous to permit stability bounds this large. For long time integrations the constant $C(T) = e^{\alpha T}$ can become very large. Assume that the consistency error is of order $\mathcal{O}(h^q)$ for some $q > 0$. The

Lax-Equivalence theorem states that the error in the numerical solution can be bound by $C(T)Th^q$. The linear growth in T is due to accumulation of the truncation error (see the proof of Theorem 3.1). We can do nothing about this. If the stability constant $C(T)$ can only be bound by (4.2), it could be very large. Consequently, we may have to excessively refine the mesh in order to make the consistency error extremely small to get a small error in the solution. Therefore, unless the PDE itself requires the possibility of exponential growth, most often one attempts to prove the stronger stability criterion (4.1), which allows only a linear growth of the error with T.

4.2 Problem Setup

A convenient setting for linear problems is that of a periodic problem. It allows us to expose basic characteristics of a discretization method, while avoiding the (important, but distinct) issue of boundary conditions.

Model Problem 2 (Periodic Heat Equation) We consider the following problem

$$\begin{aligned} u_t &= a u_{xx}, & (x,t) \in [0, 2\pi) \times (0, T], \\ u(x,0) &= \phi(x) & x \in [0, 2\pi) \\ u(x+2\pi, t) &= u(x,t), & \text{for all } x \\ \phi(x+2\pi) &= \phi(x), & \text{for all } x \end{aligned} \tag{4.3}$$

A scheme for Model Problem 2 is easily obtained by a simple modification of the explicit scheme for the heat equation with Dirichlet boundary conditions (Scheme 1):

Scheme 4 (Explicit scheme for Model Problem 2) Consider the grid

$$\mathcal{G}_h := \{ih : i = 0, \dots N-1; hN = 2\pi\}. \tag{4.4}$$

The temporal grid can be chosen as

$$\mathcal{G}_\tau := \{n\tau : n = 0, 1, \dots, M;\ M\tau = T\} \tag{4.5}$$

For $i = 0, \dots N-1$ and $n = 0, \dots M-1$ we define

$$u_i^{n+1} := \lambda u_{i-1}^n + (1-2\lambda) u_i^n + \lambda u_{i+1}^n, \tag{4.6}$$

where $\lambda := a\tau/h^2$. Periodic boundary conditions are implemented by taking $u_{i+N}^n = u_i^n$, whenever an index falls outside the permissible range. In particular we need values for $i = -1$, and $i = N$:

$$\begin{aligned} u_{-1}^n &= u_{N-1}^n, & n = 1, \dots, M \\ u_N^n &= u_0^n, & n = 1, \dots, M. \end{aligned}$$

The scheme is well defined as soon as we add the initial conditions

$$u_i^0 = u_0(ih), \qquad i = 0, \dots, N-1,$$

where $ih \in \mathcal{G}_h$, as defined in (4.4), and u_0 is a 2π-periodic function.

Define the spatial grid as in (4.4), and the temporal discretization is as in (4.5). We can define a general scheme for the abstract problem as

$$\sum_{m=-r}^{s} \widetilde{d}_m u_{i+m}^{n+1} = \sum_{m=-p}^{q} d_m u_{i+m}^{n} \qquad i = 0, \ldots, N-1, \quad n = 0, \ldots M-1, \tag{4.7}$$

where it is understood that we take any index increment "modulo N": By adding or sbtracting N, we map indices that fall outside the range $0, \ldots, N-1$ to the appropriate index established by the periodicity. The advantage of the periodic setting is that the equations are the same for all i, and no special treatment near the boundaries is necessary. Note that the indices $\widetilde{d}, d$ are functions of the mesh parameters h, τ. We collect the unknowns into a vector $\mathbf{u} = (u_0, \ldots, u_{N-1})^T$, and write the corresponding matrix form

$$\widetilde{D}_{h,\tau} \mathbf{u}^{n+1} = D_{h,\tau} \mathbf{u}^n, \tag{4.8}$$

where the periodic index shifts lead to circulant matrices. Here is an example:

Example 4.1 The Crank-Nicolson scheme (Scheme 3 defined in Section 3.2.4) can be re-written for the periodic Model Problem 4 as the task of computing

$$\begin{aligned} \frac{u_i^{n+1} - u_i^n}{\tau} &= a \left(\frac{1}{2} \frac{u_{i-1}^{n+1} - 2u_i^{n+1} + u_{i+1}^{n+1}}{h^2} + \frac{1}{2} \frac{u_{i-1}^{n} - 2u_i^{n} + u_{i+1}^{n}}{h^2} \right), \\ u_{i+N} &= u_i \end{aligned} \tag{4.9}$$

for $i = 0, \ldots, N-1$ and $n = 0, \ldots, M-1$. Using our usual definition $\lambda = a\tau/h^2$, we can re-write (4.9) as

$$-\frac{\lambda}{2} u_{i-1}^{n+1} + (1+\lambda) u_i^{n+1} - \frac{\lambda}{2} u_{i+1}^{n+1} = \frac{\lambda}{2} u_{i-1}^{n} + (1-\lambda) u_i^{n} + \frac{\lambda}{2} u_{i+1}^{n},$$

which is in the form (4.7), with

$$\begin{aligned} \widetilde{d}_0 &= (1+\lambda) \\ \widetilde{d}_{-1} = \widetilde{d}_1 &= -\frac{\lambda}{2} \\ d_0 &= (1-\lambda) \\ d_{-1} = d_1 &= \frac{\lambda}{2}. \end{aligned}$$

All other coefficients are zero. In (4.8), the matrices are given by

$$\widetilde{D}_{h,\tau} = \begin{pmatrix} (1+\lambda) & -\frac{\lambda}{2} & & & & -\frac{\lambda}{2} \\ -\frac{\lambda}{2} & (1+\lambda) & -\frac{\lambda}{2} & & & \\ & -\frac{\lambda}{2} & (1+\lambda) & -\frac{\lambda}{2} & & \\ & & & \ddots & & \\ & & & -\frac{\lambda}{2} & (1+\lambda) & -\frac{\lambda}{2} \\ -\frac{\lambda}{2} & & & & -\frac{\lambda}{2} & (1+\lambda) \end{pmatrix},$$

$$D_{h,\tau} = \begin{pmatrix} (1-\lambda) & \frac{\lambda}{2} & & & & \frac{\lambda}{2} \\ \frac{\lambda}{2} & (1-\lambda) & \frac{\lambda}{2} & & & \\ & \frac{\lambda}{2} & (1-\lambda) & \frac{\lambda}{2} & & \\ & & & \ddots & & \\ & & & \frac{\lambda}{2} & (1-\lambda) & \frac{\lambda}{2} \\ \frac{\lambda}{2} & & & & \frac{\lambda}{2} & (1-\lambda) \end{pmatrix}.$$

4.3 Stability in Maximum Norm

We can write Scheme 4 for the periodic problem as

$$\mathbf{u}^{n+1} = D_{h,\tau}\mathbf{u}^n, \tag{4.10}$$

where

$$D_{h,\tau} = \begin{pmatrix} (1-2\lambda) & \lambda & & & & \lambda \\ \lambda & (1-2\lambda) & \lambda & & & \\ & \lambda & (1-2\lambda) & \lambda & & \\ & & & \ddots & & \\ & & & \lambda & (1-2\lambda) & \lambda \\ \lambda & & & & \lambda & (1-2\lambda) \end{pmatrix}. \tag{4.11}$$

Because Scheme 4 is explicit, we have $\widetilde{D} = I$. Consider equation (4.10): In this representation the operator mapping the solution at time instance t^n to the solution at time instance t^{n+1} is given by the matrix $D_{h,\tau}$. A sufficient condition for stability in maximum norm is then

$$\|D_{h,\tau}\|_\infty \leq 1,$$

uniformly in the grid parameters h, τ. (It should be appreciated that the matrix size increases, as we let $h, \tau \to 0$, and even becomes infinite in the limit.) The norm $\|D_{h,\tau}\|_\infty$ is defined as

$$\|D_{h,\tau}\|_\infty := \max_{\mathbf{x}\neq 0} \frac{\|A\mathbf{x}\|_\infty}{\|\mathbf{x}\|_\infty}$$

There is actually an explicit formula for the maximum norm:

Theorem 4.1 *Let $A = (a_{ij})_{i,j=1,\ldots,n}$ be a square matrix. The norm $\|A\|_\infty$ is given by*

$$\|A\|_\infty = \max_{i=1,\ldots n} \sum_{j=1}^{n} |a_{ij}| \ , \tag{4.12}$$

i.e. the maximum absolute row sum.

Proof Using the definition of the norm $\|\cdot\|_\infty$ for vectors, one has

$$\|A\mathbf{x}\|_\infty = \max_{1\leq i\leq n} \left| \sum_{j=1}^{n} a_{ij}x_j \right| \leq \max_{1\leq i\leq n} \sum_{j=1}^{n} |a_{ij}x_j| \leq \|\mathbf{x}\|_\infty \max_{1\leq i\leq n} \sum_{j=1}^{n} |a_{ij}|,$$

and thus

$$\|A\|_\infty \leq \max_{1\leq i\leq n} \sum_{j=1}^{n} |a_{ij}|.$$

We show that the reverse inequality also holds, which completes the proof.

We may assume that A contains non-zero entries. (Otherwise (4.12) is trivial.) Let k be the index of a non-zero row, and let $\mathbf{x}$ be such that $x_j = 1$ whenever $a_{kj} = 0$, and $x_j = a_{kj}/|a_{kj}|$ otherwise. This construction yields $\|\mathbf{x}\|_\infty = 1$, so that

$$\|A\|_\infty \geq \|A\mathbf{x}\|_\infty = \max_{1\leq i\leq n} \left| \sum_{j=1}^{n} a_{ij}x_j \right| \geq \left| \sum_{j=1}^{n} a_{kj}x_j \right| = \sum_{j=1}^{n} |a_{kj}|.$$

We can do such a construction for *any* non-zero row k. For a zero row the same inequality holds trivially, so we can conclude

$$\|A\|_\infty \geq \max_{1\leq k\leq n} \sum_{j=1}^{n} |a_{kj}|$$

■

Using formula (4.12), clearly we have $\|D_{h,\tau}\|_\infty > 1$ if $\lambda > 1/2$. (The off-diagonal terms in (4.11) already sum to a value greater than one.) On the other hand for $0 < \lambda \leq 1/2$, we have $\|D_{h,\tau}\|_\infty = 1$. We thus get the same stability restriction that we had guessed in Section 3.5.

We can show stability in the maximum norm in a more general way. Consider a generic one-step explicit difference scheme,

$$u_i^{n+1} = \sum_{m=-p}^{q} d_m\, u_{i+m}^n. \tag{4.13}$$

It is straightforward to show that consistency requires

$$\sum_{m=-p}^{q} d_m = 1. \tag{4.14}$$

To see this, consider a Taylor's series expansion. We substitute the exact solution u (which we assume is smooth) into (4.13) and express the left-hand side with its Taylor series in time, and likewise all the terms on the right-hand side with the corresponding Taylor series in x, both centered at (x_i, t^n). For consistency, consider

$$u_i^{n+1} - \sum_{m=-p}^{q} d_m\, u_{i+m}^n = u + \tau u_t + \frac{1}{2}\tau^2 u_{tt} + \cdots - \sum_{m=-p}^{q} d_m \left(u + (mh)u_x + \frac{1}{2}(mh)^2 u_{xx} + \dots \right),$$

where u and all derivatives are evaluated at (x_i, t^n). Clearly, if the scheme is to be consistent with an equation involving no source terms (i.e. involving only the derivatives of u, not u itself), the zero-order terms must cancel, and hence (4.14) must hold. In this case, the scheme is stable in maximum norm if an only if all coefficients d_m are non-negative. To see this, it is enough notice that

$$\sum_{m=-p}^{q} |d_m| \geq \sum_{m=-p}^{q} d_m = 1,$$

with equality if and only if all coefficients are non-negative.

The stability analysis described in this section is convenient for explicit schemes. However, in general it is much more cumbersome to obtain estimates for implicit schemes, as it involves the explicit computation of the inverse matrix. The method discussed in the next section does not have that problem.

4.4 Von Neumann Stability analysis

4.4.1 General Remarks

Let the grid $\mathcal{G}_h$ be defined in (4.4), and consider the general scheme (4.7) in matrix form

$$\widetilde{D}_{h,\tau}\mathbf{u}^{n+1} = D_{h,\tau}\mathbf{u}^n. \tag{4.15}$$

Define an inner product as[1]

$$(\mathbf{u}, \mathbf{v})_{\mathcal{G}_h} := \frac{h}{2\pi} \sum_{i=0}^{N-1} u_i \overline{v}_i, \tag{4.16}$$

[1]Note carefully that we use complex arithmetic to define the inner product, although we are dealing with real vectors. (For $z = x + \hat{\imath}y \in \mathbb{C}$, recall from Appendix A.3 that $\overline{z}$ is the complex conjugate $\overline{z} = x - \hat{\imath}y$.) The reason for this is that it turns out to be convenient to choose a complex basis. This will be explained below.

Up to a scaling factor, this is the standard inner product in the Euclidean vector space. Recall that any inner product induces a corresponding norm, and we define $\|\mathbf{u}\|_{\mathcal{G}_h} := \sqrt{(\mathbf{u},\mathbf{u})_{\mathcal{G}_h}}$. Von Neumann Stability is basically stability in this norm. This means that we ought to show

$$\left\| \widetilde{D}_{h,\tau}^{-1} D_{h,\tau} \right\|_{\mathcal{G}_h} \leq 1.$$

Let us assume that with respect to the inner product (4.16) we have an orthonormal basis $\{\mathbf{v}_k\}_{k=0,\ldots,N-1}$. This is to say we can write the numerical solution as

$$\mathbf{u} = \sum_{k=0}^{N-1} c_k \mathbf{v}_k \tag{4.17}$$

for coefficients c_k that (like for any orthonormal basis) can be chosen as

$$c_k = (\mathbf{u}, \mathbf{v}_k), \tag{4.18}$$

and we may compute

$$\begin{aligned} \|\mathbf{u}\|^2 &= \left(\sum_k c_k \mathbf{v}_k, \sum_k c_k^n \mathbf{v}_k \right) \\ &= \sum_k \sum_l c_k \overline{c}_l \underbrace{(\mathbf{v}_k, \mathbf{v}_l)}_{\delta_{kl}} \\ &= \sum_k |c_k|^2. \end{aligned}$$

This is the well-known result that the norm is "in the coefficients". It holds for any orthonormal basis.

4.4.2 Choosing a Basis

We choose $\mathbf{v}_k = (v_{k,0} \ldots, v_{k,N-1})^T$ such that

$$v_{k,i} := e^{\hat{\imath} k i h}, \qquad k, i = 0, \ldots, N-1, \tag{4.19}$$

where $\hat{\imath} = \sqrt{-1}$ is the imaginary unit. These vectors form an orthonormal basis with respect to the scalar product defined in (4.16). (We will not be troubled by the fact that the basis is complex-valued. The only implication is that the coefficients determined in (4.18) will also be complex, and we will still obtain real-valued vectors in the end.)

To see that the basis is orthonormal, consider

$$\begin{aligned} (\mathbf{v}_k, \mathbf{v}_l) &= \frac{h}{2\pi} \sum_{i=0}^{N-1} e^{\hat{\imath} k i h} e^{-\hat{\imath} l i h} \\ &= \frac{h}{2\pi} \sum_{i=0}^{N-1} e^{\hat{\imath}(k-l) i h} \\ &= \frac{h}{2\pi} \sum_{i=0}^{N-1} \left(e^{\hat{\imath}(k-l)h} \right)^i, \end{aligned}$$

and define

$$q := e^{\hat{\imath}(k-l)h}.$$

Then, for $k = l$ we have $q = 1$ and $(\mathbf{v}_k, \mathbf{v}_l) = \frac{h}{2\pi} N = 1$. For $k \neq l$ we have $q \neq 1$, and we can use the geometric series

$$\sum_{i=0}^{N-1} q^i = \frac{1 - q^N}{1 - q}$$

to obtain

$$(\mathbf{v}_k, \mathbf{v}_l) = \begin{cases} \frac{1-e^{\hat{\imath}(k-l)hN}}{1-e^{\hat{\imath}(k-l)h}} & k \neq l \\ 1 & k = l \end{cases}.$$

Noting that by definition of the grid (4.4) we have $hN = 2\pi$, and thus

$$e^{\hat{\imath}(k-l)hN} = e^{\hat{\imath}(k-l)2\pi} = \cos((k-l)2\pi) + \hat{\imath}\sin((k-l)2\pi) = 1.$$

Thus we obtain

$$(\mathbf{v}_k, \mathbf{v}_l) = \delta_{kl}.$$

The main reason for choosing the complex exponential in (4.19) is the following *shift property*:

$$\begin{aligned} v_{k,i+m} &= e^{\hat{\imath}k(i+m)h} \\ &= e^{\hat{\imath}kmh}e^{\hat{\imath}kih} \\ &= \mu_k^m v_{k,i}, \end{aligned} \tag{4.20}$$

where

$$\mu_k := e^{\hat{\imath}kh}. \tag{4.21}$$

This implies that the vectors $\mathbf{v}_k$ not only form an orthonormal basis, but are actually eigenvectors of the matrices $\widetilde{D}_{h,\tau}, D_{h,\tau}$ in (4.15). Indeed, by the shift property we can write the ith component[2] of $D_{h,\tau}\mathbf{v}_k$ as

$$\sum_{m=-p}^{q} d_m v_{k,i+m} = \left(\sum_{m=-p}^{q} d_m \mu_k^m\right) v_{k,i},$$

so $D_{h,\tau}\mathbf{v}_k = \lambda_k \mathbf{v}_k$, i.e. $\mathbf{v}_k$ is the kth eigenvector with corresponding eigenvalue

$$\lambda_k = \sum_{m=-p}^{q} d_m \mu_k^m.$$

Similarly, we obtain

$$\widetilde{D}_{h,\tau}\mathbf{v}_k = \widetilde{\lambda}_k \mathbf{v}_k, \qquad \widetilde{\lambda}_k = \sum_{m=-r}^{s} \widetilde{d}_m \mu_k^m.$$

4.4.3 Stability Analysis

Using the well-known fact that, if λ is an eigenvalue of a matrix A with eigenvector $\mathbf{v}$, then λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector, we have

$$\begin{aligned} \widetilde{D}_{h,\tau}^{-1} D_{h,\tau}\mathbf{v}_k, &= \lambda_k D_{h,\tau}^{-1}\mathbf{v}_k, \\ &= \lambda_k \widetilde{\lambda}_k^{-1}\mathbf{v}_k, \end{aligned}$$

i.e. $\lambda_k\widetilde{\lambda}_k^{-1}$ is an eigenvalue of $\widetilde{D}_{h,\tau}^{-1}D_{h,\tau}$ with eigenvector $\mathbf{v}_k$. We define

$$\rho_k(h,\tau) := \frac{\lambda_k}{\widetilde{\lambda}_k} = \frac{\sum_{m=-p}^{q} d_m \mu_k^m}{\sum_{m=-r}^{s} \widetilde{d}_m \mu_k^m} \qquad k = 0, \ldots, N-1, \tag{4.22}$$

where $\rho_k(h,\tau)$ is called the amplification factor. We have emphasized that ρ depends on the grid parameters h, τ, through the coefficients $\widetilde{d}, d$.

[2]Recall that the circulant structure of the matrix implies that all components of the matrix-vector product have the same form, up to periodic index shifts of N. Do to the 2π-periodic structure of the complex exponential, the basis vectors $\mathbf{v}_k$ are invariant under such shifts: $e^{\hat{\imath}k(i+N)h} = e^{\hat{\imath}kih}e^{\hat{\imath}kNh} = e^{\hat{\imath}kih}$. (Recall: $hN = 2\pi$.) For practical purposes, we may thus ignore such modifications near the boundary, when operating on v_{i+m}.

Theorem 4.2 *Let $\rho_k(h,\tau)$ be defined as in (4.22), and let $\widetilde{D}_{h,\tau}, D_{h,\tau}$ be the circulant matrices corresponding to the coefficients $\widetilde{d}, d$ in (4.22). Then*

$$\left\|\widetilde{D}_{h,\tau}^{-1} D_{h,\tau}\right\|_{\mathcal{G}_h} = \max_{0\le k\le N-1} |\rho_k(h,\tau)| \tag{4.23}$$

Proof Consider arbitrary $\mathbf{u} \in \mathbb{R}^N$. Taking the norm, we have

$$\begin{aligned}
\left\|\widetilde{D}^{-1} D\mathbf{u}\right\|_{\mathcal{G}_h} &= \left\|\sum_{k=0}^{N-1} c_k \widetilde{D}^{-1} D\mathbf{v}_k\right\|_{\mathcal{G}_h} \\
&= \left\|\sum_{k=0}^{N-1} c_k \lambda_k \widetilde{D}^{-1} \mathbf{v}_k\right\|_{\mathcal{G}_h} \\
&= \left\|\sum_{k=0}^{N-1} c_k \frac{\lambda_k}{\widetilde{\lambda}_k} \mathbf{v}_k\right\|_{\mathcal{G}_h} \\
&\le \max_{0\le k\le N-1} |\rho_k(h,\tau)| \left\|\sum_{k=0}^{N-1} c_k \mathbf{v}_k\right\|_{\mathcal{G}_h} \\
&= \max_{0\le k\le N-1} |\rho_k(h,\tau)| \, \|\mathbf{u}\|_{\mathcal{G}_h} .
\end{aligned}$$

This holds for arbitrary $\mathbf{u}$. Hence

$$\sup_{\mathbf{u}\neq 0} \frac{\left\|\widetilde{D}^{-1} D\mathbf{u}\right\|_{\mathcal{G}_h}}{\|\mathbf{u}\|_{\mathcal{G}_h}} = \left\|\widetilde{D}^{-1} D\right\|_{\mathcal{G}_h} \le \max_{0\le k\le N-1} |\rho_k(h,\tau)| . \tag{4.24}$$

To show the desired equality, we merely have to show that equality in (4.24) is achieved for some vector $\mathbf{u}$. Now let k_0 be such that

$$\max_{0\le k\le N-1} |\rho_k(h,\tau)| = |\rho_{k_0}(h,\tau)| ,$$

and consider the vector $\mathbf{u} = \mathbf{v}_{k_0}$. Then,

$$\begin{aligned}
\left\|\widetilde{D}^{-1} D\mathbf{v}_{k_0}\right\|_{\mathcal{G}_h} &= \left\|\widetilde{D}^{-1} \lambda_k \mathbf{v}_{k_0}\right\|_{\mathcal{G}_h} \\
&= \left\|\widetilde{\lambda}_{k_0}^{-1} \lambda_{k_0} \mathbf{v}_{k_0}\right\|_{\mathcal{G}_h}
\end{aligned}$$

Noting $\widetilde{\lambda}_{k_0}^{-1} \lambda_{k_0} = \rho_{k_0}(h,\tau)$, we conclude

$$\frac{\left\|\widetilde{D}^{-1} D\mathbf{v}_{k_0}\right\|_{\mathcal{G}_h}}{\|\mathbf{v}_{k_0}\|_{\mathcal{G}_h}} = |\rho_{k_0}(h,\tau)| .$$

Equality in eq. (4.24) is achieved, and eq. (4.23) follows. ∎

The amplification factor is a computable expression, and to put a bound on it is an (often cumbersome) algebraic exercise. In practice we verify that $\max_k |\rho_k(h,\tau)| \le 1$ uniformly in the grid parameters h, τ. Then we can conclude that the scheme is stable.

Example 4.2 Consider the explicit method for the heat equation:

$$u_i^{n+1} = \lambda u_{i-1}^n + (1-2\lambda)u_i^n + \lambda u_{i+1}^n,$$

where again $\lambda = \frac{a\tau}{h^2}$. Comparing to the generic form (4.7) one has

$$\begin{aligned} \widetilde{d}_0 &= 1 \\ d_0 &= 1 - 2\lambda \\ d_{-1} = d_1 &= \lambda, \end{aligned}$$

and all other coefficients are zero. Substituting this into the general formula (4.22) yields

$$\rho_k = \left|\lambda\mu_k^{-1} + (1-2\lambda) + \lambda\mu_k\right| = \left|\lambda\left(e^{-\hat{\imath}kh} + e^{\hat{\imath}kh}\right) + 1 - 2\lambda\right| \leq 1$$

using

$$e^{-\hat{\imath}kh} + e^{\hat{\imath}kh} = 2\cos(kh) \tag{4.25}$$

we obtain

$$-1 \leq 1 + 2\lambda \underbrace{(\cos(\xi) - 1)}_{\leq 0} \leq 1,$$

where we have defined $\xi := kh$. (Note that we need show the inequality for any $h > 0$. Hence we have to let the product kh assume any value in $[0, 2\pi)$, so that it is easier to just use a generic variable ξ.) The right inequality holds because $\cos(\xi) \leq 1$ for all arguments. For the left inequality, $\cos(\xi) \leq 1$ yields

$$\begin{aligned} & -1 \leq 2\lambda(-1-1) + 1 \\ \Leftrightarrow \quad & -1 \leq -4\lambda + 1 \\ \Leftrightarrow \quad & \lambda \leq \frac{1}{2} \end{aligned}$$

Hence we satisfy the Neumann condition if we choose $\lambda \leq 1/2$, and we obtain the familiar result for this example.

Note that formula (4.22) need not be "remembered". It is easily re-derived by simply applying both sides of the numerical scheme to one basis function $\mathbf{v}_k$ evaluated at node i:

Example 4.3 Implicit method for the heat equation:

$$-\lambda u_{i-1}^{n+1} + (1+2\lambda)u_i^{n+1} - \lambda u_{i+1}^{n+1} = u_i^n,$$

with $\lambda = \frac{a\tau}{h^2}$. Apply both sides to the basis $\mathbf{v}_k = e^{\hat{\imath}kih}$ and take the ratio. This yields

$$\rho_k = \frac{e^{\hat{\imath}kih}}{-\lambda e^{\hat{\imath}k(i-1)h} + (1+2\lambda)e^{\hat{\imath}kih} - \lambda e^{\hat{\imath}k(i+1)h}}$$

Applying standard rules of exponentiation, one has

$$\rho_k = \frac{1}{-\lambda e^{-\hat{\imath}kh} + (1+2\lambda) - \lambda e^{\hat{\imath}kh}}$$

And again using identity (4.25) for the complex exponential, we have

$$|\rho_k| = \left| \frac{1}{1 + 2\lambda(\underbrace{1 - \cos(kh)}_{\geq 0})} \right| \leq 1 \tag{4.26}$$

We see that the amplification factor is always less than unity, regardless of the choice of λ, i.e. independently of the grid parameters. We thus have no time step restriction, and the scheme is unconditionally stable.